Exercise 3

On ratio between two consecutive terms in the binomial expansion

From the school book | Understand | Apply | Higher Order Thinking Skills

1 Choose the correct answer from those given :

(1) In the expansion of : $(x + y)^{10}$ according to the descending exponents of x , the ninth term : the eighth term equals .....

(a) $\frac{3y}{8x}$ (b) $\frac{3x}{8y}$ (c) $\frac{8y}{3x}$ (d) $\frac{8x}{3y}$

(2) In the expansion of : $(1 - x)^{12}$ according to the ascending exponents of x , the coefficient of sixth term : the coefficient of fifth term equals .....

(a) $\frac{8}{5}$ (b) $\frac{5}{8}$ (c) $\frac{-8}{5}$ (d) $\frac{-5}{8}$

(3) In the expansion of : $(3 + a x)^9$ according to the ascending exponents of x , if the coefficient of T_3 = the coefficient of T_4 then a = ...

(a) $\frac{9}{5}$ (b) $\frac{9}{4}$ (c) $\frac{9}{7}$ (d) $\frac{9}{2}$

(4) If $T_{17} = T_{18}$ in the expansion of $(2 + x)^{50}$ according to the ascending exponents of x , then x = ...

(a) 1 (b) 2 (c) 3 (d) 4

(5) In the expansion of : $(x + y)^8$ according to the descending exponents of x , the ratio $T_6 : T_4$ equals .....

(a) $25y^2 : 16x^2$ (b) $25x^2 : 16y^2$ (c) 1 (d) $y^2 : x^2$

(6) In the expansion of $(3a - 2b)^{11}$ according to the descending exponents of a , if the ratio between the two middle terms respectively equals $\frac{-3}{2}$, then a : b = ...

(a) 9 : 4 (b) 4 : 9 (c) 1 (d) – 1

(7) In the expansion of : $(a - b)^n$ according to the descending exponents of a where $n \geq 5$, if T_5 , T_6 each is additive inverse of the other then $\frac{a}{b}$ = ...

(a) $\frac{n-5}{6}$ (b) $\frac{n-4}{5}$ (c) $\frac{5}{n-4}$ (d) $\frac{6}{n-5}$

(8) The ratio between the two middle terms in the expansion of $\left(x+\frac{1}{x}\right)^{2n-1}$ according to the descending exponents of x respectively = ...

(a) x^2 (b) $\frac{x^2}{2n}$ (c) $\frac{x^2}{n}$ (d) $\frac{2n+1}{2n+3}$

(9) In the expansion of $\left(\frac{1}{x}-x^2\right)^9$ according to the ascending exponents of x , if $T_5 = 12\,T_4$, then x = ...

(a) 2 (b) – 2 (c) 8 (d) – 8

(10) ***(Trial 2023)*** In the expansion of $x^n\left(\frac{1}{x}-2\right)^n$ according to the ascending exponents of x , if $T_5 = 2016\,x^4$, then the ratio between the coefficient of T_3 and the coefficient of T_4 equals .....

(a) 3 : 7 (b) – 3 : 14 (c) – 1 : 14 (d) – 3 : 7

(11) The two consecutive terms in the expansion of $(3 + 2x)^{74}$ according to the ascending exponents of x that have equal coefficients are .....

(a) T_{29} , T_{30} (b) T_{30} , T_{31} (c) T_{31} , T_{32} (d) T_{28} , T_{29}

(12) In the expansion of $(1 + x)^{2n}$ according to the ascending exponents of x , if the coefficient of T_3 is the arithmetic mean between the coefficients of T_2 , T_4 , then $2n^2 - 9n$ = ...

(a) – 8 (b) – 6 (c) – 7 (d) – 18

(13) If the ratio between the coefficients of two consecutive terms in the expansion of $(1 + x)^{24}$ according to the ascending exponents of x is 4 : 1 then the two terms are .....

(a) T_4 , T_5 (b) T_{20} , T_{21} (c) T_3 , T_4 (d) T_{21} , T_{22}

(14) The term which have the greatest coefficient in the expansion of $(1 + x)^{10}$ according to the ascending exponents of x is .....

(a) T_{11} (b) T_5 (c) T_6 (d) T_{10}

(15) The term which has the smallest coefficient in the expansion of $(2x + 7y)^3$ according to the descending exponents of x is .....

(a) T_4 (b) T_3 (c) T_2 (d) T_1

(16) In the expansion of $(x + y)^n$ according to the descending exponents of x if the seventh term is the term of greatest coefficient , then n = ...

(a) 12 (b) 13 (c) 14 (d) 15

(17) The greatest coefficient in the expansion of $(2x + y)^8$ equals .....

(a) 1120 (b) 448 (c) 1792 (d) 1028

(18) If the sum of the coefficients of all terms of the expansion $(1 + 2x)^n$ according to the ascending exponents of x equals 6561 , then the greatest coefficient in the expansion equals .....

(a) 896 (b) 3594 (c) 1792 (d) 1972

(19) The greatest term in the expansion of $(1 + 4x)^8$ according to the ascending exponents of x at $x = \frac{1}{3}$ equals .....

(a) $56 \times \left(\frac{4}{3}\right)^5$ (b) $56\left(\frac{3}{4}\right)^5$ (c) $56\left(\frac{3}{4}\right)^4$ (d) $56\left(\frac{2}{5}\right)^4$

(20) If the greatest coefficient in the exponsion of $(a + x)^{20}$ is the coefficient of T_{11} , then $a \in$ ..... where $a \in \mathbb{R}^+$

(a) $\left[\frac{10}{11} , \frac{11}{10}\right]$ (b) $[10 , 11]$ (c) $\left[\frac{11}{10} , \frac{10}{11}\right]$ (d) $\left[\frac{-9}{11} , \frac{10}{11}\right]$

2 Find the ratio between the fourth term and the third term in the expansion of : $(3 + x)^8$ according to the ascending exponents of x « $\frac{2}{3}x$ »

3 Find the ratio between the third term and the fifth term in the expansion of : $(x - 2)^7$ according to the descending exponents of x « $\frac{3x^2}{20}$ »

4 Find the ratio between the coefficient of the ninth term and the coefficient of the tenth term in the expansion : $(2x - 3y)^{14}$ according to the descending exponents of x « -1 »

5 Find the ratio between the seventh term and the eighth in term in the expansion of : $\left(x + \frac{1}{3}x^{-1}\right)^{15}$ according to the descending exponents of x when $x = \frac{1}{2}$ « $\frac{7}{12}$ »

6 In the expansion of : $\left(x^2 + \frac{2}{x}\right)^8$ according to the descending exponents of x

(1) Find the ratio between the fifth and the sixth term , and if this ratio equals 8 : 25 , find the value of x

(2) Prove that this expansion does not have a term free of x « $\frac{5x^3}{8}$, $\frac{4}{5}$ »

7 *(2nd Session 2014)* In the expansion of : $(1 + 2x)^n$ according to the ascending exponents of x , if the ratio between the eighth term and the tenth term is 1 : 28 when $x = 3$ find the value of n « 15 »

8 *(2nd Session 2012)* Find the order and the value of the term free of x in the expansion of : $(x^3 - \frac{1}{x})^8$ according to the descending exponents of x , then find the ratio between the middle term in this expansion and the next term to the middle term when $x = 2$

« T_7 , 28 , – 20 »

---

9 If the middle term of the expansion of : $(1 + x)^{10}$ according to the ascending exponents of x is equal to twice the seventh term , find the value of x « $\frac{3}{5}$ »

---

10 If the ratio between the term free of x and the previous term to it in the expansion of : $(x + \frac{1}{2x})^{12}$ according to the descending exponents of x is 7 : 3 find the value of x « $\pm\frac{1}{2}$ »

---

11 In the expansion of : $(x^2 - \frac{1}{x})^{14}$ according to the descending exponents of x , prove that there is no term free of x , then find the ratio between the seventh term and the sixth term when $x = -1$ « $\frac{3}{2}$ »

---

12 In the expansion of : $(x^2 + \frac{1}{x})^9$ according to the descending exponents of x , find the term free of x , and if the ratio between the term free of x and the sixth term is 9 : 4 , find the real value of x « 84 , $\frac{2}{3}$ »

---

13 In the expansion of : $(5 + 4x)^{16}$ according to the ascending exponents of x , if one of the terms of the expansion equals three times the next term for it , find the order of these two terms if $x = 1$ « T_{12} , T_{13} »

---

14 Find the greatest term in the expansion of : $(3 + 2x)^6$ according to the ascending exponents of x at $x = 1$ « T_3 »

---

15 *(1st Session 2013)* In the expansion of : $(2 + \frac{x}{3})^9$

(1) Find the value of x which makes the two middle terms equal.

(2) Without finding the expansion , find the value of the greatest coefficient. « 6 , 768 »

---

16 In the expansion of : $(1 + x)^8$ according to the ascending exponents of x , if $2T_5 = T_4 + T_6$, find the value of x « 2 or $\frac{1}{2}$ »

---

17 In the expansion of : $(2x - 3)^{15}$ according to the descending exponents of x , find the values of x which make $13T_3 + 10T_4 + T_5 = 0$ « $\frac{9}{2}$, $\frac{1}{2}$ »

18 In the expansion of : $\left(\frac{3x}{2} + \frac{2}{3x^2}\right)^{12}$, find each of the middle term and the term containing x^{-3} , and if the ratio between these two term equals 7 : 4 , then find the value of x

« $\frac{2}{3}$ »

19 In the expansion of : $(x + y)^8$ according to the descending exponents of x , if the ratio between T_3 , T_4 equals 1 : 4 , and the middle term is 1120 , then find each of x , y

« ± 1 , ± 2 »

20 In the expansion of : $(3 + 2x)^n$ according to the ascending exponents of x , if $T_5 = 6\ T_4$, $T_8 = 3\ T_7$, then find each of n , x

« 27 , $\frac{3}{2}$ »

21 ***(1st Session 2017)*** In the expansion of : $(1 + x)^n$ according to the ascending exponents of x , if $T_3 = 17$, $3\ T_2 \times T_4 = 544$, then find the value of each : n , x

« 18 , $\pm\frac{1}{3}$ »

22 In the expansion of : $\left(2x + \frac{3}{x^2}\right)^n$ according to the descending exponents of x , if $T_9 = T_{10}$ and $\frac{T_6}{T_7} = \frac{8}{15}$, then find the value of n , and prove that the expansion has no term free of x

« n = 20 »

23 In the expansion of : $(1 + x)^n$ according to the ascending exponents of x , if $(T_3)^2 = T_2 \times T_5$, find the value of n when $x = \frac{9}{5}$

« 7 »

24 In the expansion of : $(1 + x)^n$ according to the ascending exponents of x , if the coefficient of T_{10} is the arithmetic mean between the coefficient of T_9 , and the coefficient of T_{11} , find each of :

(1) n **(2)** ${}^nP_3$ **(3)** $\underline{|n - 19}$

« 23 or 14 , 10626 or 2184 , 24 »

25 ***(2nd Session 2013)*** In the expansion of : $(1 + x)^n$ according to the ascending exponents of x , if $T_5 = T_6$ and $T_4 = \frac{25}{3}\ T_2$, then find the value of each of n , x

« 10 , $\frac{5}{6}$ »

26 In the expansion of : $(x + y)^n$ according to the descending exponents of x , if the second term is the arithmetic mean between the first term and the third term at $x = 2y$, find the value of n.

« 8 »

27 ***(1st Session 2012)*** In the expansion of : $\left(x^2 + \frac{1}{2x}\right)^{3n}$ according to the descending exponents of x :

(a) Prove that the term free of x is of order $(2n + 1)$

(b) Find the ratio between the term free of x and the middle term at n = 4 and $x = 1$

« $\frac{15}{112}$ »

28 If the ratio among the fifth , sixth and seventh terms in the expansion of : $\left(\frac{3x}{2}+\frac{2}{3x}\right)^n$ according to the descending exponents of x is 40 : 24 : 11 , find the value of n , x « 16 , $\pm\frac{4}{3}$ »

29 If the ratio among the coefficients of three consecutive terms in the expansion of : $\left(x+\frac{k}{x^2}\right)^{27}$ according to descending exponents of x is 15 : 6 : 2
where $k \in \mathbb{Z}^+$, find the orders of these terms , then find the order and the value of the term free of x in this expansion. « T_{20} , T_{21} , T_{22} »

30 ***(2nd Session 2001)*** If the coefficients of three consecutive terms in the expansion of : $(1+x)^n$ are 35 , 21 , 7 according to the ascending exponents of x find the value of n and the order of the three terms. « 7 , T_5 , T_6 , T_7 »

31 In the expansion of : $(a+b)^n$ according to the descending exponents of a , if the second , the third and the fourth terms are 240 , 720 , 1080 respectively , find the values of a , b and n « 2 , 3 , 5 »

32 If the coefficients of the fourth , fifth and sixth terms in the expansion : $(2x+y)^n$ according to the descending exponents of x form an arithmetic sequance , find the value of n « 19 or 8 »

33 In the expansion of : $(1+x)^n$ according to the ascending exponents of x if $3\,T_4$, $\sqrt{5}\,T_6$ and $6\,T_8$ are in geometrical sequence , then find the value of n « 12 »

34 In the expansion of : $\left(\sqrt{x}+\frac{1}{x}\right)^8$, if T_4 , T_5 , $25\,T_7$, T_6 are proportional , find the value of x « $\frac{5}{2}$ »

35 ***(2nd Session 2008)*** Prove that the expansion of : $\left(x^2+\frac{1}{x}\right)^n$ containing a term free of x if n is a multiples of 3 , and if n = 12 and the expansion according to the descending exponents of x , find the ratio between the term free of x and the coefficient of the next term for it. « $\frac{9}{4}$ »

36 In the expansion of : $\left(2x+\frac{3}{x^2}\right)^n$ according to the descending exponents of x , if the ninth and the tenth terms are equal and the ratio between the sixth and the seventh terms is 8 : 15 , find the value of n and prove that the expansion has no term free of x « 20 »

37 In the expansion of : $(1 + m\,x)^n$ according to the ascending exponents of x , if $4\,T_6 = 7\,T_8$, $\frac{T_4}{T_6} = \frac{1}{4}$ when $x = 1$, find the value of m , n « ± 2 , 8 »

38 If $T_2 : T_3$ in the expansion of : $(a + b)^n$ according to the descending exponents of a equals the ratio $T_3 : T_4$ in the expansion of $(a + b)^{n+3}$ according to the descending exponents of a , find the value of n « 5 »

39 If the order of the term free of x in the expansion of : $\left(2x^2 - \frac{3}{x}\right)^{21}$ according to the descending exponents of x is equal to the order of the term free of x in the expansion of : $\left(x + \frac{1}{x}\right)^{2n}$ according to the descending exponents of x , find the value of n , then find the ratio between the two middle terms of the first expansion at $x = -1$ respectively. « 14 , $\frac{2}{3}$ »

Higher skills

40 **Choose the correct answer from those given :**

(1) In the expansion of : $(2 + x)^9$ according to descending exponents of x if the sixth term added to $\frac{1}{4}$ of the sevenths term equals 7 times the eighth term then $x =$ .....

(a) 3 or $\frac{8}{3}$ (b) -3 or $\frac{-8}{3}$ (c) 3 or $\frac{-8}{3}$ (d) -3 or $\frac{8}{3}$

(2) In the expansion of : $(4 - x)^8$ according to the ascending exponents of x , if T_5 is the greatest numerical value then $x \in$ .....

(a) $[-5 , 5]$ (b) $\left]\frac{-16}{5} , \frac{16}{5}\right[$

(c) $[-5 , 5] - \left[-\frac{16}{5} , \frac{16}{5}\right]$ (d) $\left[-5 , \frac{-16}{5}\right] \cup \left[\frac{16}{5} , 5\right]$

41 Three consecutive terms have been taken in the expansion of : $(1 + x)^{30}$ according to the ascending exponents of x , such that the ratio between the sum of the coefficients of the first and the second terms from these three terms and the sum of the coefficients of the second and the third terms is 5 : 3 , find these three terms. « T_{19} , T_{20} , T_{21} »